

Assignment 10

Poisson Process

1. Suppose a certain bank knows from its past experience that between 11 a.m to 12 noon on each day the mean arrival rate of customer at withdrawal window is 60. Find the probability that (i) exactly 2 customers (ii) atmost 2 customers will arrive in a given one minute interval between 11 a.m and 12 noon. (Ans: 0.1839, 0.9197)
2. Suppose that the average number of telephone calls arriving at a police station is 30 calls per hour. What is the probability that (i) no calls will arrive in a 3 minute period. (ii) more than 4 calls will arrive in 5 minutes period? (Ans: 0.2231, 0.1088)
3. The number of telephone calls arriving at a certain switch board within a time interval of length measured in minutes is a poisson process $X(t)$ with parameter $\lambda = 2$. find the probability that no telephone calls arrive in 5 minutes time. (Ans: 4.54×10^{-5})
4. Customers arrive at a bank according to poisson process with mean rate of 3 customers per minute. Find the probability that during a time interval of 2 minutes (i) exactly 4 customers arrive. (ii) four or more than 4 customers arrive. (Ans: 0.133, 0.8488)
5. The arrivals of customers at a courier in a railway station is a poisson process with mean rate of 3 customers per minute. Find the probability that the interval of time between two consecutive customers is (i) more than one minute (ii) between 1 and 2 minutes (iii) 3 or less than 3 minutes. (Ans: 0.0498, 0.0473, 0.9999)
6. A radioactive substance emits particles at the rate of 6 particles per minute according to a poisson process. The probability that the emitted particle will be recorded is 0.5. find the probability that 10 particles will be recorded in 4 minutes period. (Ans: 0.104)
7. Queries presented in a computer database follow a poisson process with rate $\lambda = 6$ queries per minute. And experiment consists of monitoring the database for m minutes and recording the number of queries presented. (i) what is the probability there is no query in one minute interval (ii) what is the probability that exactly 6 queries arriving in one minute interval (iii) what is the probability that less than 3 queries arrive in half minute interval (Ans: 0.00248, 0.1607, 0.4231)
8. VLSI chips, essential for running a computer system, fail according to Poisson distribution with the rate of 1 chip in 5 weeks. If there are two spare chips in hand and if a new supply is to arrive in 8 weeks, what is the probability that during the next 8 weeks, the system will be down for a week or more owing to the lack of chips. (Ans:0.1665)
9. The rate of failure of a hard disk in a computer system follows a poisson distribution with mean rate of 1 per week. Find the probability that 2 weeks have elapsed since the last failure. If we have 5 extra hard disks and next supply is not due in 10 weeks,

find the probability that the machine will not be out of use in the next 10 weeks.(Ans: 0.135, 0.068)

10. A fisher man catches fish a poisson rate of 2 per hour from a large pond with lots of fish. If he starts fishing at 10 a.m what is the probability that he catches one fish by 10.30 a.m and three fish by noon.(Ans: 0.3679, 0.1953)